EXCEL LEARNING HANDBOOK

Lessons 9-11 | Data Formatting, Copy & Paste Special, Freeze & Unfreeze Panes

A step-by-step reference based on the lessons covered after the first Excel PDF. Includes explanations, hands-on examples, menu navigation, expected results, shortcuts, and limitations. Practice assignments and quiz questions are intentionally excluded.

Contents

Lesson	Subtopics
9. Data Formatting	Wrap Text; font and fill; headers; horizontal and vertical alignment; Format Painter (single/multiple use)
10. Copy, Paste & Paste Special	Normal paste; Paste Values; pictures; linked pictures; formulas and number formats; No Borders; Skip Blanks; Multiply, Add, Subtract, Divide
11. Freeze & Unfreeze Panes	Top row; first column; multiple columns; simultaneous row/column freeze; unfreeze; gridlines; common pitfalls

Important: Lesson 12 (Merge & Unmerge Cells) was introduced as the next lesson but has not yet been taught or practised, so it is not included.

LESSON 9 - Data Formatting

Formatting improves readability without changing underlying cell values. The goal is to create clear headers, consistent alignment, readable fonts, and reusable formatting.

9.1 Prepare a sample customer table

Customer ID	Customer Name	Email Address	City	Purchase
C101	Barry Allen	barry@gmail.com	Bangalore	750
C102	Darren Hoffman	darren@yahoo.com	Mumbai	1200
C103	Jessica Walsh	jessica@hotmail.com	Delhi	950
C104	Christine Anderson	christine@gmail.com	Chennai	1800
C105	Emily Blunt	emily@yahoo.com	Pune	650
C106	Rahul Sharma	rahul@gmail.com	Hyderabad	1450

Create worksheet 09_DataFormatting, paste headers into A1:E1 and data into A2:E7.

9.2 Remove unnecessary Wrap Text

1. Select A1:E7.

2. On the Home tab, in Alignment, check Wrap Text; click to turn it off if enabled.

3. Select rows 1:7 and choose Home > Format > AutoFit Row Height to restore appropriate row heights.

Important: Disabling Wrap Text affects display, not data. The row height does not always reset automatically; AutoFit Row Height may be necessary.

9.3 Select and emphasize the header row

1. Select A1, then Ctrl + Shift + Right Arrow to select adjacent filled headers through E1 (or select A1:E1 directly).

2. Press Ctrl + B or click Home > Bold.

3. Use Home > Fill Color to choose a light fill.

4. Use Home > Font Color to choose contrasting dark text.

Expected result: A clearly differentiated header, without making body rows difficult to read. Light backgrounds with dark text generally improve contrast; the transcript emphasizes keeping body rows readable.

9.4 Horizontal and vertical alignment

Alignment	What it does
Align Left / Center / Align Right	Positions content horizontally in a cell.
Top Align / Middle Align / Bottom Align	Positions content vertically in a cell.

1. Select A2:E7.

2. From Home > Alignment, select Center for horizontal centering.

3. Select Middle Align for vertical centering.

4. Observe that values move within cells but do not change.

Important: For many analysis tables, left-aligned names and right-aligned amounts can be easier to scan than centering every column.

9.5 Font family, size, emphasis and colors

1. Select A2:E7 and choose Home > Font > Calibri (or your preferred workbook font).

2. Set font size to 11.

3. Experiment with Bold (Ctrl+B), Italic (Ctrl+I), Underline (Ctrl+U), Font Color and Fill Color.

4. Restore a consistent, readable style before continuing.

9.6 Format Painter - copy formats without values

Format Painter transfers font, fill, alignment, borders, and number formatting from one range to another; it does not overwrite the destination values.

1. Format B2 with Calibri 12, bold, a light fill, dark font, and center alignment.

2. Select B2 and click Home > Clipboard > Format Painter once.

3. Click B3. The styling changes, while the value Darren Hoffman remains unchanged.

Single use: One click on Format Painter applies the style once and then turns off.

1. Select B2 again and double-click Format Painter.

2. Click B4, B5, B6 and B7 to apply formatting repeatedly.

3. Press Esc or click Format Painter again to stop.

Range example: Format A2:E2, select that range, click Format Painter and drag across A3:E7. Values remain in place, but formatting is replicated.

Important: Format Painter is for presentation, not for copying actual table contents. Matching source/destination range shapes generally gives the intended pattern.

9.7 Formatting quick reference

Action	Windows shortcut or menu
Bold	Ctrl+B
Italic	Ctrl+I
Underline	Ctrl+U
Fill Color / Font Color	Home > Font
Horizontal / vertical alignment	Home > Alignment
Wrap Text	Home > Alignment > Wrap Text
AutoFit Row Height	Home > Format > AutoFit Row Height
Format Painter	Home > Clipboard > Format Painter

LESSON 10 - Copy, Paste & Paste Special

Normal paste copies more than visible values: typically it brings formulas and source formatting. Paste Special lets you decide exactly what to transfer or which calculation to perform.

10.1 Normal copy-and-paste

Employee ID	Name	Department	Salary
E101	Barry Allen	IT	45000
E102	Darren Hoffman	HR	38000
E103	Jessica Walsh	Finance	52000
E104	Emily Blunt	IT	48000
E105	Rahul Sharma	Sales	42000

1. On worksheet 10_CopyPaste, enter the employee table in A1:D6.

2. Make source headers bold, add a light fill and apply All Borders.

3. Give destination range F1:I6 a different fill and borders.

4. Copy A1:D6 (Ctrl+C), click F1, and paste (Ctrl+V).

Expected result: Content, source-cell formatting, borders and any formulas are pasted; pre-existing destination formatting can be replaced.

10.2 Paste Values - copy calculated output instead of formulas

Example in A10:C14: Basic Salary, Bonus, Total Salary. Enter the first formula in C11, then fill down.

Basic Salary	Bonus	Total Salary (formula result)
40000	5000	45000
35000	3000	38000
50000	7000	57000
45000	4000	49000

C11 formula: **=A11+B11**

1. Copy C11:C14; normal-paste to E11.

2. Observe formulas may adjust their relative references (for example, =A11+B11 becomes =C11+D11 when moved two columns right).

3. Copy C11:C14 again and select G11.

4. Choose Home > Paste > Paste Special > Values; click OK.

5. Select G11: the Formula Bar displays the number 45000, not the formula.

Important: Shortcut for Paste Special: Ctrl+Alt+V. In the classic dialog, V chooses Values, then Enter confirms.

10.3 Copy as Picture - fixed visual snapshot

1. Select A1:D6.

2. Choose Home > Copy dropdown > Copy as Picture.

3. Choose Appearance: As shown on screen; Format: Picture; click OK.

4. Paste into an empty area. The result is an image object, not editable table cells.

5. Repeat using Appearance: As shown when printed and compare rendering.

Expected result: A static picture for Word, PowerPoint, or reports. Changing a source value later does not update the picture.

10.4 Linked Picture - visual that follows source changes

1. Copy A1:D6 as cells using Ctrl+C.

2. Choose an empty destination and Home > Paste dropdown > Linked Picture (name/location may vary by Excel version).

3. Change the source name Barry Allen to Barry A. Allen and press Enter.

4. Observe that the linked picture updates, while the picture itself does not allow direct editing of its underlying cells.

Important: A linked picture depends on its source. When copying/sharing workbooks, verify that references remain available.

10.5 Formulas and Number Formats

1. Apply Currency or Accounting number formatting to salary calculation cells.

2. Copy the formulas.

3. At a destination, open Paste Special and choose Formulas and Number Formats.

4. Verify formulas and currency/number formats were copied without decorative fonts, background fills, or border styling.

Important: Relative formula references can still shift after pasting. Check results at the destination.

10.6 Paste without source borders

1. Copy the bordered employee table A1:D6.

2. Choose Home > Paste dropdown > No Borders at a clear destination.

3. Compare with a normal paste: contents and much of the source appearance carry over, but source borders do not.

Paste option	Retained content
Values	Final cell results; not original formulas or source presentation styles.
Formulas and Number Formats	Formulas plus number formatting; no decorative formatting.
No Borders	Contents and most source formats; not copied source borders.

LESSON 10 - Advanced Paste Special (continued)

10.7 Skip Blanks - update only revised bonuses

Create an employee bonus table in A20:C25. Leave source revision cells genuinely empty when there is no change.

Employee	Old Bonus	Revised Bonus
Barry Allen	5000	6500
Darren Hoffman	4000	(blank)
Jessica Walsh	6000	7500
Emily Blunt	4500	(blank)
Rahul Sharma	5500	6000

1. Select revised values C21:C25 and copy.

2. Select B21, the top of Old Bonus.

3. Open Paste Special (Ctrl+Alt+V).

4. Under Paste choose Values and tick Skip blanks.

5. Confirm with OK.

Employee	Old Bonus after update
Barry Allen	6500
Darren Hoffman	4000
Jessica Walsh	7500
Emily Blunt	4500
Rahul Sharma	6000

Important: The blanks must actually be empty. A source cell with a formula returning an empty string is not the same as a genuinely blank cell.

10.8 Paste Special operations - arithmetic without helper formulas

Paste Special offers None, Add, Subtract, Multiply and Divide. Choose Values to avoid carrying number-formatting from the operation cell. These actions change the destination cells directly.

10.8.1 Multiply: increase prices by 14%

An increase of 14% means multiply by 114% = 1.14, not multiply by 114.

Product	Old price	Calculation	New price
Pizza	750	750 x 1.14	855
Burger	500	500 x 1.14	570

Product	Old price	Calculation	New price
Fries	300	300 x 1.14	342
Sandwich	400	400 x 1.14	456

1. Enter the old prices in a separate range and place 114% into an unused cell (for example I20).

2. Copy the 114% cell.

3. Select only the old-price numeric cells.

4. Open Paste Special; under Paste select Values and under Operation select Multiply.

5. Click OK and verify 750 becomes 855.

10.8.2 Add: increase prices by a fixed 114

Use a fresh copy of the original prices, not the already multiplied results.

1. Type 114 into a helper cell (numeric value, not percentage) and copy it.

2. Select the original price range.

3. Open Paste Special > Values; under Operation select Add.

4. Click OK. Expected: 750 becomes 864; 500 becomes 614; 300 becomes 414; 400 becomes 514.

10.8.3 Subtract and Divide

Operation	Helper value	Expected example
Subtract	50	750 - 50 = 700
Divide	2	750 / 2 = 375

For either operation: copy the numeric helper cell, select target values, open Paste Special, select Values, choose Subtract or Divide and confirm.

10.9 Choosing the correct paste option

Need	Recommended option
Exact duplicate including cell style	Normal Paste
Final result without a formula	Paste Values
A fixed image of a table	Copy as Picture
An image that refreshes when data changes	Linked Picture
Formulas and numeric formats, not font/fill	Formulas and Number Formats
Contents without copied borders	No Borders
Update only filled source rows	Values + Skip blanks
One-time bulk price increase	Values + Multiply (e.g. 114%)
One-time fixed monetary increase	Values + Add

Important: For destructive one-time operations, save a copy of original values first. Copy/Paste shortcuts: Ctrl+C, Ctrl+V, Ctrl+X, Ctrl+Alt+V; Undo: Ctrl+Z.

LESSON 11 - Freeze & Unfreeze Panes

Freeze Panes keeps identifying rows or columns visible while you scroll large datasets. It changes only the worksheet view; no cell data is changed.

11.1 Prepare a wide customer dataset

Create worksheet 11_FreezePanes. Use these 12 headers in A1:L1, then populate at least 30 rows (repetition is fine for scrolling practice).

Columns	Headers
A-D	Customer ID | Customer Name | City | State
E-H	Country | Age | Product | Order Amount
I-L	Payment Method | Order Status | Acquisition Source | Campaign

Sample row: C101 | Barry Allen | Bangalore | Karnataka | India | 28 | Laptop | 55000 | Credit Card | Delivered | Google Ads | PROMO2020.

1. Enter the 12-column headers and at least several sample customer rows.

2. Repeat sample rows to reach approximately 30-40 records.

3. Select the table and choose Home > Format > AutoFit Column Width.

4. Bold the header row to make it easily distinguishable.

11.2 Freeze Top Row

1. Scroll down and observe that the header disappears.

2. Return near the top and choose View > Freeze Panes > Freeze Top Row.

3. Scroll to row 30. Row 1 stays visible; the rest of the worksheet scrolls.

The thin boundary below row 1 shows where the frozen region ends.

11.3 Unfreeze Panes

1. Choose View > Freeze Panes > Unfreeze Panes.

2. Scroll down and verify that the header moves out of view again.

11.4 Freeze First Column

1. Choose View > Freeze Panes > Freeze First Column.

2. Scroll horizontally to the campaign field in column L.

3. Column A (Customer ID) remains visible; column B is not frozen.

11.5 Freeze multiple columns

1. Unfreeze any existing panes.

2. Select cell C1 - immediately to the right of columns A and B.

3. Choose View > Freeze Panes > Freeze Panes.

4. Scroll right: columns A and B stay visible; row 1 scrolls vertically because no rows were frozen.

11.6 Freeze header row and first two columns together

1. First choose Unfreeze Panes.

2. Select cell C2.

3. Choose View > Freeze Panes > Freeze Panes.

4. Scroll down and right. Row 1 and columns A:B stay on screen.

Rule: General Freeze Panes freezes the rows above the selected cell and columns to its left. C2 therefore freezes row 1 and columns A:B.

11.7 Which cell should you select?

Keep visible	Select cell
Row 1 only (general Freeze Panes)	A2
Columns A:B only	C1
Row 1 and column A	B2
Row 1 and columns A:B	C2
Rows 1:2 only	A3
Rows 1:2 and columns A:B	C3
Rows 1:3 and columns A:C	D4
Rows 1:4 and columns A:B	C5

11.8 Gridlines and freeze boundaries

1. Choose View > Show > Gridlines to switch gridline visibility off.

2. Observe that the freeze boundary can be easier to notice.

3. Turn Gridlines on again if preferred.

Important: Gridlines are worksheet guides, not manually inserted borders. Turning them off does not remove data.

11.9 Common mistakes

Issue	Explanation / correction
C2 vs B2	C2 freezes columns A:B; B2 freezes column A only (along with row 1).
Using Freeze Top Row after another freeze	This replaces the previous configuration rather than adding to it. Unfreeze, then use C2 + Freeze Panes to combine.
Existing frozen panes	Use Unfreeze Panes before making a different setup.
Freeze Panes vs Split	Freeze keeps chosen rows/columns visible; Split creates independently scrollable worksheet panes.

11.10 Quick reference

Action	Excel navigation
Freeze top row	View > Freeze Panes > Freeze Top Row
Freeze first column	View > Freeze Panes > Freeze First Column
Freeze rows + columns	Select boundary cell, then View > Freeze Panes > Freeze Panes
Remove freeze	View > Freeze Panes > Unfreeze Panes
Show/hide gridlines	View > Show > Gridlines

Final Reference - Lessons 9 to 11

Formatting: Build legible headers, align consistently, and copy styling with Format Painter without overwriting values.

Copy/Paste Special: Select the correct paste behavior for formulas, values, pictures, blanks, and one-time math operations; verify that formulas and linked objects still work.

Freeze Panes: Select a cell immediately below and right of everything you want to keep visible, then choose Freeze Panes.

This volume covers the topics practised after the initial Lessons 1-8 handbook and ends with Lesson 11. The next lesson in the learning sequence is Lesson 12, Merge & Unmerge Cells.